

Persistence and Predictability in Multi-station Soil Moisture Forecasting from the USDA SCAN Sensor Network

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Abstract. Soil moisture forecasting supports irrigation scheduling and drought early warning, yet machine-learning studies in this area rarely compare their models against the simplest available predictor: carrying the current sensor reading forward. Using 708,000 hourly records from 22 USDA SCAN stations in the Southern Plains between 2023 and 2026, this work examines how much predictive skill learned models add over this persistence baseline for soil moisture at 8-inch depth, at horizons of 24 and 72 hours. Four tabular models and a two-layer LSTM are trained to predict the change in moisture rather than its level, so that any improvement can be read directly as learned skill. Persistence proves a demanding reference, with a mean absolute error of 0.629 percentage points at 24 hours. In the main benchmark all four tabular models fall short of it, by 4 to 44 percent, while the LSTM reduces error by 13.8 percent at 24 hours and 11.7 percent at 72 hours; a separately tuned LightGBM model reaches parity but not more. Paired tests on more than 90,000 held-out samples confirm both outcomes at $p < 0.001$, indicating that hand-crafted lag features add limited incremental information over a sequential representation when consecutive observations are nearly identical. The accuracy gap does not, however, prevent practical use: a threshold-based drought alert built on a model that loses to persistence still recovers 97.8 percent of drought episodes, because alerting depends on the direction of a forecast rather than its precision. Meteorological variables contribute a further 7.4 percent, and an apparent gain on unseen stations is shown to reflect site-level variability rather than genuine transfer.

Keywords: soil moisture forecasting · persistence baseline · LSTM · drought early warning · USDA SCAN · precision agriculture

1 Introduction

Soil moisture governs crop water availability, irrigation timing, and the onset of agricultural drought. Sensor networks such as the USDA Natural Resources Conservation Service Soil Climate Analysis Network (SCAN) report hourly multi-depth moisture alongside meteorological observations at hundreds of sites across the United States, and a forecast one to three days ahead is exactly the lead

time a farmer needs to schedule irrigation before crops experience water stress [3]. The literature reflects this demand, with gradient-boosted trees, recurrent networks, and transfer learning on satellite products all reporting encouraging accuracy on the task [1,2,5].

Persistence, however, is not consistently reported as a reference baseline, which makes the incremental value of learned models difficult to interpret across studies. Soil moisture at depth behaves as a storage variable: water entering the profile drains and evaporates over days rather than hours, so the value observed now is already a strong estimate of the value a day later. Lorenz et al. [4] describe the consequence for forecasting, noting that as the autocorrelation of a target approaches unity the optimal predictor converges towards persistence, leaving progressively less room for a learned model to improve upon it. An error of 0.7 percentage points appears small until one discovers that carrying the current reading forward achieves 0.63 on the same data.

This work therefore asks how much skill learned models add over persistence for soil moisture at 8-inch depth, and which families of model add any at all. The question is examined on 22 SCAN stations in Oklahoma, Texas, and Kansas over 2023 to 2026, comprising 708,000 hourly observations, at horizons of 24 and 72 hours. Every model is trained to predict the change in moisture rather than its absolute level, so that a model predicting no change reproduces the persistence baseline exactly and any departure from it can be attributed to learned skill. The outcome divides sharply along a single axis, and not the one the tabular-versus-deep-learning literature would suggest [8]: in the main benchmark all four tabular models perform worse than persistence, while a modest two-layer LSTM improves upon it at both horizons.

2 Related Work

The SCAN network was established to provide ground reference measurements of soil moisture, soil temperature, precipitation and air temperature at hourly resolution [3]. It has been widely used to validate satellite retrievals but remains comparatively underused as a forecasting benchmark, which is part of what motivates the present study.

Several recent studies apply machine learning to soil moisture prediction from sensor or satellite sources. Alahmad et al. [2] compared Random Forest, XGBoost, and LSTM for multi-depth prediction from IoT sensors and reported comparable accuracy across the three, with the sequential model marginally ahead, although the evaluation covered a single season at one location. Widyasuti et al. [1] applied transfer learning to satellite-derived moisture across Tasmania and demonstrated that pre-trained models can generalise spatially, though their work operated on gridded products rather than point sensors. Basak et al. [6] are closest to the present study, contrasting autoregressive moving average models with LSTM variants at depths of 5, 15, and 30 centimetres and observing that the benefit of model complexity diminishes as sensor depth increases. Fang and Shen [5] introduced a data-integration LSTM for satellite moisture and

showed that sequential architectures exploit temporal structure beyond what hand-crafted lag features provide. Across this literature, however, persistence is not consistently reported as a reference baseline, which makes the magnitude of the improvements claimed in different studies difficult to compare.

A parallel line of work concerns how models should be trained when the target is strongly autocorrelated. Lorenz et al. [4] argue that forecasting the change in a variable rather than its level separates the trivial component of the prediction from the learnable residual, and prevents a model from receiving credit for reproducing its own input. The present work adopts this formulation for all models and uses it as the basis of evaluation rather than as a training convenience.

The choice of model family is also contested. Zeng et al. [7] showed that simple linear models remain competitive with Transformer architectures on series with strong trend components, while Grinsztajn et al. [8] argued that tree ensembles continue to outperform deep learning on typical tabular data. These findings motivate including linear, bagged, and boosted baselines alongside a sequential model. The results reported here delimit their scope: when the available features are lagged copies of the target itself, the tabular advantage reverses.

3 Methodology

The task is to predict soil moisture at 8-inch depth, denoted `sm8` and measured in percent, at a horizon of 24 or 72 hours ahead of the current hour. Every learned model is trained to predict the change in moisture over that horizon rather than its absolute value, with the final forecast obtained by adding the predicted change to the current reading. This arrangement means that a model predicting zero change reproduces the persistence forecast exactly, so an improvement over persistence measures learned skill and a deficit indicates that the model has actively degraded a signal it was given.

Twenty features are constructed from the raw hourly records using SQL window functions partitioned by station and ordered by timestamp, which guarantees that no future value enters any row. Moisture is included at all four available depths, and the target depth additionally contributes lags at 1, 6, 24, and 168 hours, chosen to span short-term response, the diurnal cycle, and the weekly scale, together with two change features formed by differencing against the 6-hour and 24-hour lags. Meteorological conditions enter through hourly precipitation, 24-hour and 7-day rainfall accumulations, air temperature, a 3-day mean temperature, and wind speed. Soil temperature at 2 inches, the hour of day, the month, and an integer station code complete the set.

Four tabular models span the usual complexity spectrum. Ridge regression with L2 regularisation represents the linear case [9], Random Forest with 300 trees the bagged case [10], and LightGBM and XGBoost the boosted case, both configured with 800 estimators, a learning rate of 0.03, and 80 percent subsampling [11,12]. LightGBM is additionally tuned by randomised search over 30 configurations with blocked time-series cross-validation [13]. The sequential

model is a two-layer LSTM operating on a 24-hour lookback window, with 64 and 32 hidden units separated by dropout, a small dense layer, and a linear output predicting the change in moisture. Training uses the Adam optimiser with mean absolute error loss and early stopping on a validation split. Importantly, the LSTM receives the same twenty features as the tabular models; only their arrangement differs, from a single row to a sequence of 24 consecutive hours.

Performance is reported as mean absolute error in percentage points, averaged over five random seeds for the tabular models and three for the LSTM, whose training cost is higher. Because seed variance measures optimisation stability rather than predictive uncertainty, paired tests are additionally run on the per-sample error series of each pair of models. Training covers observations before 1 January 2026 and testing those from that date onward, a strict chronological split that mirrors the operational setting in which a model is fitted on history and deployed forward. For comparisons involving the LSTM, the first 24 test rows are dropped from every model so that all methods are scored on identical samples.

Three further analyses use LightGBM as the representative tabular model, chosen because tree-based attribution supports exact Shapley values [14] and because its training cost permits repeated fitting. A leave-stations-out protocol holds four of the 22 stations entirely out of training; an ablation removes all six meteorological features simultaneously; and a drought alert is triggered whenever predicted moisture falls below 11 percent, targeting the operational threshold of 10 percent with a one-point buffer for forecast error.

4 Experimental Setup

Hourly records covering January 2023 to September 2026 were retrieved from the USDA NRCS Air and Water Database for 22 SCAN stations in the Southern Plains [15]. Reshaping the wide export into long format yields 707,520 station-hours, of which 608,379 remain after removing rows without a forecastable target. Moisture readings outside the physical range of 0 to 60 percent were treated as missing, and gaps of at most two hours were linearly interpolated within each station. Longer gaps were deliberately left unfilled, since interpolating across them would manufacture smooth trajectories and inflate the very persistence baseline the study sets out to measure. Missing rates rise with depth, reaching 8.2 percent at 20 inches and 5.5 percent at 4 inches, compared with under 1.5 percent for the meteorological variables, which reflects the practical difficulty of maintaining deep sensors in the field.

The network is not homogeneous. The distribution of sm8 across all stations is clearly bimodal, with modes near 10 and 40 percent corresponding to arid sites in West Texas and humid sites in the east of the region, which is why station identity is included among the features. Figure 1 shows three representative regimes, in which drought episodes appear as slow declines and rainfall recharge as sharp rises. Seasonality is statistically significant but modest in amplitude,

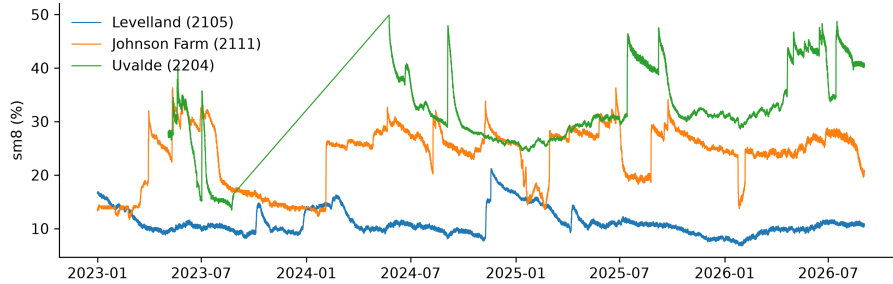


Fig. 1. Soil moisture at three representative stations between 2023 and 2026. The distinct baseline levels across sites motivate including a station encoding, while the smoothness of each trajectory between rainfall events is what makes persistence difficult to improve upon.

Table 1. Chronological split and the persistence benchmark for each horizon.

Horizon	Train	Test	Persistence MAE (%)
24 hours	397,882	95,030	0.629
72 hours	397,279	93,798	1.381

with monthly means ranging from 20.7 percent in October, the driest month, to 27.1 percent in February.

Far more consequential for forecasting is the structure within each series rather than across them. The correlation between sm8 and its value 24 hours earlier exceeds 0.99 at most stations, which is the central fact of this study. Adjacent depths are also strongly correlated, while instantaneous precipitation correlates only weakly with moisture and contributes mainly through its accumulated forms.

Table 1 reports the resulting split together with the error each horizon must beat. The figure of 0.629 percentage points at 24 hours sets a demanding bar, since a model whose predictions deviate from the current reading by more than roughly six tenths of a point on average will lose to doing nothing at all. All experiments ran in Python on a standard workstation processor without graphics acceleration, which is deliberate given that the intended deployment context is an agricultural extension office rather than a compute cluster.

5 Results

5.1 Model Comparison Across Horizons

Table 2 and Figure 2 present the central result for the main benchmark, in which every model uses its default configuration. The LSTM improves on persistence by 13.8 percent at 24 hours and 11.7 percent at 72 hours, while all four tabular

Table 2. Test-set mean absolute error in percentage points, averaged over five seeds for tabular models and three for the LSTM. Negative improvement values indicate that the model performs worse than persistence.

Model	MAE (24 h)	Improvement	MAE (72 h)	Improvement
Persistence	0.629	—	1.381	—
Ridge	0.654	−4.0%	1.436	−4.0%
Random Forest	0.830	−32.0%	1.985	−43.7%
LightGBM	0.669	−6.4%	1.526	−10.5%
XGBoost	0.715	−13.7%	1.581	−14.5%
LSTM	0.542	+13.8%	1.220	+11.7%

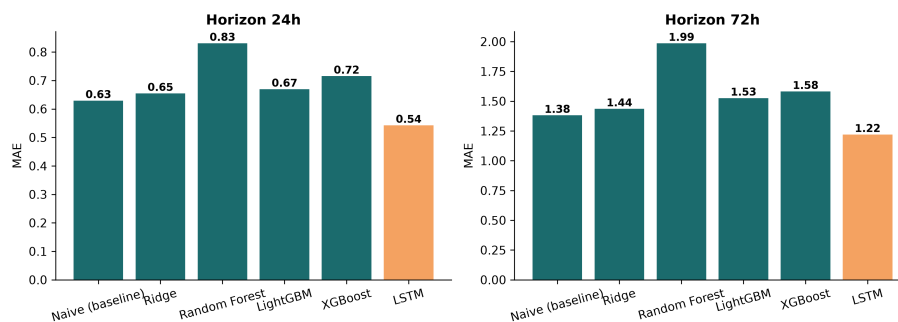


Fig. 2. Mean absolute error by model at 24 hours (left) and 72 hours (right) for the main benchmark. The LSTM is the only learned model that improves on persistence at both horizons; a separately tuned LightGBM model reaches near-parity at 24 hours.

models perform worse than the baseline. Ridge falls short by 4.0 percent at both horizons, LightGBM by 6.4 and 10.5 percent, XGBoost by 13.7 and 14.5 percent, and Random Forest by 32.0 and 43.7 percent.

Seed standard deviations reach at most 0.014, so the ordering itself is stable, but several gaps are narrow enough to warrant a direct test. Paired comparisons on the per-sample errors place the LSTM advantage well beyond plausible sampling variation at both horizons ($p < 0.001$). The same tests show that the tabular deficit is statistically significant rather than incidental: across test sets of more than 90,000 samples, Ridge and LightGBM consistently produce higher absolute errors than persistence rather than merely failing to improve upon it.

Tuning substantially narrows the gap at the 24-hour horizon. Randomised search reduces LightGBM error from 0.682 to 0.625, compared with 0.629 for persistence, a change roughly seven times the seed-to-seed standard deviation and therefore not attributable to random initialisation. The selected configuration favours heavier regularisation than the default, with a lower learning rate, fewer leaves, and more estimators, which is consistent with the model being penalised for variance around a nearly constant target. This places the tuned model essentially at parity with the baseline, although the margin is small and comes

from a single tuned run rather than from the multi-seed protocol used for the main benchmark. A carefully tuned boosted tree can therefore match persistence at this horizon, but the improvement observed for the LSTM remains an order of magnitude larger.

5.2 Error Structure

The residuals of the LightGBM model are informative rather than pathological. Training error of 0.46 sits below test error of 0.68, indicating mild but not severe overfitting. Errors are unevenly distributed across the year: April is the worst month at 1.11 against an annual average of 0.68, while summer months fall to between 0.5 and 0.7. April corresponds to the spring wetting and drying transition, when the smooth autoregressive behaviour that dominates the remainder of the year temporarily breaks down. Errors are also elevated in the wettest decile of the target distribution, where rainfall recharge produces the sharp rises visible in Figure 1.

5.3 Feature Attribution

Shapley attribution on the LightGBM model, shown in Figure 3, confirms the structural reading of the main result. Current moisture dominates by a wide margin, followed by station identity, surface moisture, and 24-hour rainfall accumulation. The lagged moisture terms that were intended to carry temporal information rank below all of these.

This ordering explains the pattern in Table 2. A tree ensemble given four lags of a variable whose consecutive hourly values differ by fractions of a percentage point receives four near-copies of its own input. The marginal information they carry is small, while the variance they introduce is not, and the model spends capacity fitting noise. The LSTM, handed the same twenty variables arranged as a sequence of 24 steps, can instead represent the shape of the recent trajectory rather than four isolated samples from it.

Two of the attribution patterns are physically interpretable. Station identity ranking second reflects the bimodal distribution described earlier, indicating that the model is largely learning site baselines rather than dynamics. The dependence on the 24-hour lag is negatively sloped, meaning that high lagged moisture pushes the predicted change downward. This pattern is consistent with mean-reverting moisture dynamics, in which wet soil drains and dry soil recovers, but the learned relationship does not translate into a meaningful improvement over persistence.

5.4 Meteorological Contribution

Removing all six meteorological features at once raises the LightGBM error from 0.682 to 0.733, a degradation of 7.4 percent. The contribution is real but modest, and the reason is visible in the attribution ranking. Rainfall influence is already encoded within the autoregressive moisture terms, since past precipitation is

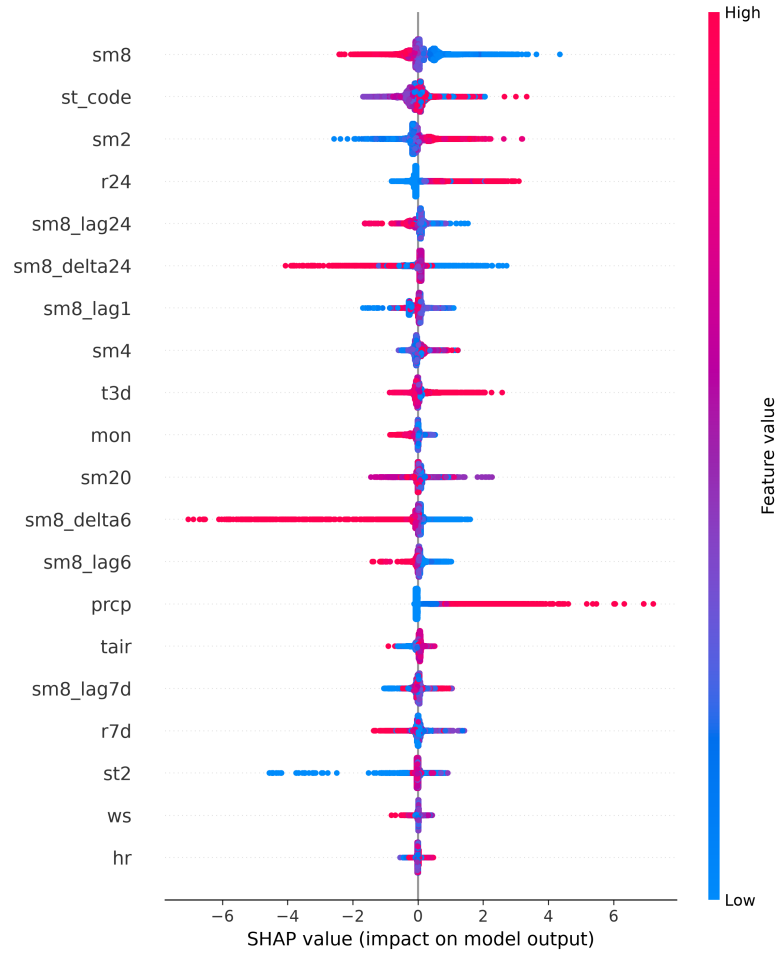


Fig. 3. Shapley attribution for LightGBM at the 24-hour horizon. Current moisture dominates the prediction, the station code enters second by absorbing site-specific baselines, and the lag features intended to encode dynamics rank below both.

Table 3. Leave-stations-out evaluation at the 24-hour horizon. The lower error on unseen stations reflects their lower moisture variability rather than stronger generalisation, as the persistence baseline falls by a similar proportion.

Setting	MAE (%)
LightGBM, all stations	0.682
LightGBM, four unseen stations	0.332
Persistence, four unseen stations	0.268

precisely what produced the current moisture level. Weather features therefore add information mainly about events that have not yet propagated to the 20-centimetre layer.

For deployment this represents a useful economy, suggesting that a single weather station serving several moisture nodes is a reasonable design and that budget is better spent on moisture sensor density than on co-located meteorological instrumentation.

5.5 Transfer to Unseen Stations

Holding four of the 22 stations entirely out of training produces, at first glance, a striking outcome. Error on the unseen stations is 0.332, well below the 0.682 obtained by the full model, which read naively would suggest that the model generalises better to sites it has never encountered.

Table 3 suggests otherwise. Persistence on the same four stations scores 0.268, also far below its network-wide value of 0.629. The apparent improvement is therefore partly explained by the held-out sites exhibiting lower moisture variability than the network average, which lowers every method’s error proportionally. Measured against the correct local baseline the transferred model still loses, by approximately the same margin as it loses network-wide, so the result is consistent with the main finding rather than an exception to it. A single random partition cannot separate generalisation from site composition, and a more defensible evaluation would require repeated leave-out folds stratified by moisture regime.

5.6 Drought Alerting

If a model loses to persistence on average error, the question arises whether it retains any practical value. For the application that motivates this work the answer is affirmative. A drought alert that fires whenever predicted moisture falls below 11 percent, targeting the operational threshold of 10 percent, recovers 97.8 percent of true drought episodes at a precision of 80.5 percent, using the same LightGBM model that falls short of persistence on mean error.

Table 4 gives the full breakdown. Recall of 97.8 percent means that fewer than three drought episodes in a hundred go undetected, which suits the asymmetric cost structure of agriculture, where a missed drought may cause irreversible

Table 4. Drought alert performance at the 24-hour horizon. An alert fires when predicted moisture falls below 11 percent; a true episode is an actual reading below 10 percent.

True positives	12,241	Recall	97.78%
False positives	2,963	Precision	80.51%
False negatives	278	F1 score	88.31%
True negatives	79,548		

crop loss while a false alarm costs one unnecessary irrigation cycle. The reason a model that loses on average error performs well here is structural: a threshold alert depends on whether a forecast crosses the operational boundary rather than on minimising continuous-valued error, and the model’s errors are concentrated in the wettest decile of conditions, far from the drought threshold, where they rarely change that verdict.

6 Discussion

The failure of the tabular models is not a failure of gradient boosting as such but a consequence of the target’s autocorrelation. Lorenz et al. [4] predict that as correlation approaches unity the optimal predictor converges towards persistence, and at values above 0.99 this task sits deep within that regime. The tabular models receive lag features that are near-duplicates of the current reading, so the information they add is small while the variance they introduce is not. Training on the change rather than the level makes this visible, because the residual to be learned has a mean near zero and a tree ensemble fitted to it largely fits noise. The advantage of the LSTM arises from its inductive bias rather than from additional data, since it receives the same twenty variables and differs only in being able to represent trajectory shape. This aligns with the observations of Fang and Shen [5] and extends the depth finding of Basak et al. [6]: at 20 centimetres the tabular representations evaluated here do not merely fail to improve on persistence but measurably increase test-set error. The result also bounds the scope of the finding by Grinsztajn et al. [8] that trees outperform deep learning on tabular data, suggesting that the relative advantage of tree-based models depends on whether features carry independent information or, as here, lagged copies of the target.

Three practical implications follow. Persistence should be reported as a default baseline in soil moisture forecasting rather than treated as optional, since without it an error of 0.67 reads as success rather than as a regression of six percent against a free alternative. For forecasting deep layers at short horizons, additional sensor coverage is likely a better investment than model sophistication, given that persistence on a dense network will outperform a boosted tree on a sparse one. And model selection should follow the decision rather than the metric, as the drought alert demonstrates that a model can be operationally effective and metrically inferior at the same time.

Several limitations bound these conclusions. The 22 stations occupy a single semi-arid to sub-humid climate zone, and colder, tropical, or irrigated regions may exhibit different persistence characteristics. Only the 20-centimetre depth is evaluated, whereas surface moisture would weaken persistence and likely favour learned models while deeper layers would strengthen it further. The four-year window may exclude multi-year drought regimes in which persistence assumptions break down. Only one sequential architecture is tested, leaving Transformer and state-space models unexplored, and traditional statistical baselines such as ARIMA and ARMAX are not included, so the comparison runs between persistence and machine learning rather than across all three families. Finally, the transfer analysis rests on a single random partition, which is why it is presented diagnostically rather than as a claim about generalisation.

7 Conclusion

This work examined how much predictive skill machine-learning models add over persistence for soil moisture at 8-inch depth, and found the answer to depend on how the inputs were represented rather than on model capacity. Across 22 SCAN stations and 708,000 hourly records, a two-layer LSTM improved on persistence by 13.8 percent at 24 hours and 11.7 percent at 72 hours, while Ridge, Random Forest, LightGBM, and XGBoost all performed worse than the baseline by margins between 4 and 44 percent under their default configurations; tuning brought LightGBM to parity but no further. Given the same feature set, the difference lay in whether the model received a single feature vector or a 24-hour sequence.

Two further findings temper and extend this result. Meteorological covariates contributed only 7.4 percent, because past weather is already encoded within current moisture through physical coupling. And a drought alert built on the LightGBM model still achieved 97.8 percent recall at 80.5 percent precision despite that model’s sub-persistence error, because threshold decisions depend on which side of the operational boundary a prediction falls rather than on continuous-valued error alone.

Taken together, these results suggest that persistence deserves to be the reference point in this domain, and that within this benchmark the improvement over it came from a representation preserving the recent temporal sequence rather than treating lagged observations as an unordered feature vector. Future work will add ARIMA and ARMAX baselines to complete the comparison across model families, extend the benchmark across additional depths and climate regions, and replace the single-split transfer test with stratified repeated folds. Applying the pipeline to emerging agricultural sensor networks in Vietnam is a natural next step, with direct relevance to United Nations Sustainable Development Goals 2 and 11.

The SCAN data used here are in the public domain. The feature pipeline, seeded training scripts, and evaluation code will be released upon publication.

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